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A Face Recognition Based Attendance System with Geolocation and Real-Time Action Logging --Manuscript Draft--

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Abstract:	This paper introduces a cutting-edge face recognition-based attendance system, designed to address the limitations of traditional attendance methods through the integration of advanced machine learning, computer vision, and geospatial APIs. The system streamlines the attendance process by automating the identification and logging of attendees with high accuracy and efficiency. Key features include live video recognition for real-time face identification, an intuitive user registration module for enrolling new individuals, CSV-based logging for seamless data export and management, and geolocation-aware attendance tracking to ensure that records are not only time-accurate but also location-specific. This geospatial context provides valuable insights, particularly for distributed teams or multi-location setups. The implementation leverages Python, a versatile programming language, and integrates OpenCV for real-time video processing and face detection, ensuring quick and reliable face recognition even in dynamic environments. The graphical user interface (GUI) is developed using PyQt5, allowing for a user-friendly and responsive experience. This combination of powerful technologies ensures that the system is both scalable and adaptable, able to integrate easily into various organizational workflows, from small educational institutions to large-scale corporate environments. The system's practical application is validated through experimental results conducted in diverse settings, including workplaces, academic institutions, and security-sensitive environments. These results highlight the system's exceptional accuracy, even under challenging conditions such as low lighting or crowded spaces. Furthermore, the system demonstrates its potential to enhance operational efficiency, reduce administrative overhead, and improve security by providing a reliable, context-aware solution for attendance management. In conclusion, this face recognition-based attendance system offers a modern, automated solution that combines the power of machine learning and computer vision with geospatial data, creating an intelligent, highly effective tool for attendance tracking across a wide range of industries and applications.
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Ambika Pote
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To: Ambika Pote <ambika.pote@springernature.com>
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Subject: Re: Fw: :: Review Request: A Geolocation-Enabled Face Recognition Attendance System with Real-Time Action Logging

Dear Ambika,

Thank you for acknowledging my request. As suggested, please submit it in the special edition of SN Computer Science Journal on "Trends in Quantum Computing and Emerging Business Technologies"

Yours sincerely,
Debmalya Ray

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Looking forward for your response.

Best Regards,

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Response to Reviewers:	Dear Sir, Please find below the revised work based on the suggestion : - given attention on the problem statement and our contribution - Related works - drawn a comparison between existing approaches and proposed system with real-time logging Table Defined - Figure 2 and Figure 3 depicts the LBP and HOG techniques with actual faces - Pseudocode for the algorithm defined for both location validation and attendance mechanism - Experimental Results with actual metrics are defined in more details with Processing Time equation , Throughput equation used and equation related to geolocation accuracy.

- | | |
|--|--|
| | <ul style="list-style-type: none">- Table 3 depicting the performance based on conditions, FPS , etc of the system for computer vision system- The Github code is available on request mentioned in the Data Availability section to strengthen the solution in details- Use of Nominatim geocoder from the geopy library is been mentioned within the sub section while explaining the psuedocode that is used to validate locaton. |
|--|--|

A Face Recognition Based Attendance System with Geolocation and Real-Time Action Logging

Debmalya Ray¹

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Abstract. This paper introduces a cutting-edge face recognition-based attendance system, designed to address the limitations of traditional attendance methods through the integration of advanced machine learning, computer vision, and geospatial APIs. The system streamlines the attendance process by automating the identification and logging of attendees with high accuracy and efficiency. Key features include live video recognition for real-time face identification, an intuitive user registration module for enrolling new individuals, CSV-based logging for seamless data export and management, and geolocation-aware attendance tracking to ensure that records are not only time-accurate but also location-specific. This geospatial context provides valuable insights, particularly for distributed teams or multi-location setups.

The implementation leverages Python, a versatile programming language, and integrates OpenCV for real-time video processing and face detection, ensuring quick and reliable face recognition even in dynamic environments. The graphical user interface (GUI) is developed using PyQt5, allowing for a user-friendly and responsive experience. This combination of powerful technologies ensures that the system is both scalable and adaptable, able to integrate easily into various organizational workflows, from small educational institutions to large-scale corporate environments. The system's practical application is validated through experimental results conducted in diverse settings, including workplaces, academic institutions, and security-sensitive environments. These results highlight the system's exceptional accuracy, even under challenging conditions such as low lighting or crowded spaces. Furthermore, the system demonstrates its potential to enhance operational efficiency, reduce administrative overhead, and improve security by providing a reliable, context-aware solution for attendance management.

In conclusion, this face recognition-based attendance system offers a modern, automated solution that combines the power of machine learning and computer vision with geospatial data, creating an intelligent, highly effective tool for attendance tracking across a wide range of industries and applications.

Keywords: Face recognition, attendance system, geolocation, real-time logging, machine learning, computer vision, OpenCV, PyQt5

1. INTRODUCTION

Face recognition systems have gained significant importance in various sectors, including surveillance, attendance management, access control, and security systems. The growing reliance on these systems can be attributed to their ability to provide automated, accurate, and secure solutions. Unlike traditional methods for attendance tracking, such as manual registers or RFID-based systems, face recognition technology offers a contactless approach that not only minimizes human error but also enhances operational efficiency and reduces administrative workload. These traditional methods, though still widely used, are prone to several drawbacks, including inefficiency in large-scale environments, issues with data accuracy, and vulnerability to various security threats. RFID systems, for example, are susceptible to card duplication or loss, and manual tracking is often time-consuming, labour-intensive, and prone to fraud or human oversight [1, 2].

Furthermore, conventional attendance systems typically rely on physical devices for identity verification, which can often result in inefficiencies, especially when managing large numbers of people or employees. These systems may require individuals to swipe cards or sign in manually, which is both time-consuming and prone to error, especially in fast-paced environments [4, 5]. Face recognition systems, by contrast, are capable of automatically verifying identity without the need for direct physical interaction. This not only makes them faster and more efficient but also improves the accuracy and reliability of attendance data, reducing the possibility of errors or intentional fraud. In the context of attendance management, face recognition has emerged as a promising solution, effectively eliminating the need for cumbersome physical verification devices or manual tracking methods. The system can scan and recognize faces within seconds, making it ideal for environments with high foot traffic, such as offices, schools, and large organizations. Moreover, the integration of machine learning and computer vision techniques can significantly improve the performance and adaptability of these systems, making them capable of continuously learning and improving over time. Coupled with the inclusion of geolocation services, these systems can go beyond just facial identification and provide contextual, real-time attendance logging [6, 7].

Geospatial data integration is particularly important for ensuring that attendance records are not only accurate but also location-specific. This is especially valuable in contexts where employees or students are working remotely or need to check in from multiple locations. For instance, employees working at different sites or from home can still have their attendance tracked accurately with a system that logs their geographical location alongside their facial data, making the system more versatile and adaptable to modern work environments [8, 9]. Additionally, incorporating timestamping further guarantees that attendance records are time-sensitive and provides a robust audit trail. This research introduces a Python-based real-time face recognition and attendance system that leverages advanced face encoding methods in conjunction with geolocation APIs. The face recognition component utilizes state-of-the-art algorithms for precise and rapid identification, ensuring that attendance records are accurate and reliable [10]. Simultaneously, the system's integration with geolocation services ensures that each attendance entry is tagged with key contextual data, such as location coordinates and timestamps, offering a complete and accurate record of attendance in real time [11].

1.1 Problem Statement

Traditional attendance systems encounter several critical challenges. RFID systems are vulnerable to proxy attacks, with a reported 12% fraud rate. Fingerprint scanners raise hygiene concerns, particularly in post-pandemic settings. Manual registers suffer from a 30% error rate due to human oversight.

1.2 Our Contribution

To address these issues, we have developed a novel attendance system that incorporates hybrid feature extraction, combining HOG (Histogram of Oriented Gradients) for structural features and LBP (Local Binary Patterns) for texture details. This combination enhances robust recognition under varying lighting conditions. Additionally, our system employs geospatial validation to ensure that attendance is marked only from authorized locations, utilizing reverse geocoding through the Nominatim API.

This combination of cutting-edge face recognition technology and geospatial awareness presents an intelligent alternative to traditional attendance systems, solving many of the problems associated with outdated methods. The implementation of automation within this system also reduces the administrative burden often involved in manual attendance tracking, freeing up valuable resources that can be better utilized elsewhere in an organization. Additionally, the system is designed with scalability and flexibility in mind, making it easily adaptable to various organizational contexts, including educational institutions, corporate environments, and even remote workforce management [12, 13]. By leveraging this system, organizations can improve operational efficiency, ensure data accuracy, and bolster security, all while eliminating the risk of human error and fraud commonly associated with traditional attendance tracking systems [14].

2. RELATED WORK

Numerous approaches have been proposed for attendance systems using biometrics and facial recognition. Prior studies focus on:

2.1 Biometric Fingerprint Attendance Systems

Biometric fingerprint attendance systems utilize the unique patterns found in an individual's fingerprints to verify identity. These systems are highly accurate and reliable, as fingerprints are unique to each person and remain unchanged over time. However, despite their effectiveness, there are several challenges associated with their use. One significant concern is **intrusiveness**—the need for physical contact with a fingerprint scanner can be perceived as invasive, particularly in environments where hygiene is a priority, such as schools, offices, or healthcare facilities [10]. In addition, **hygiene concerns** arise when multiple individuals use the same fingerprint scanner, as these devices can harbour bacteria and viruses, potentially spreading infections among users. These issues have spurred interest in the development of **contactless biometric systems**, which aim to reduce the need for physical interaction while maintaining the accuracy and security of traditional fingerprint-based systems.

2.2 RFID-Based Solutions

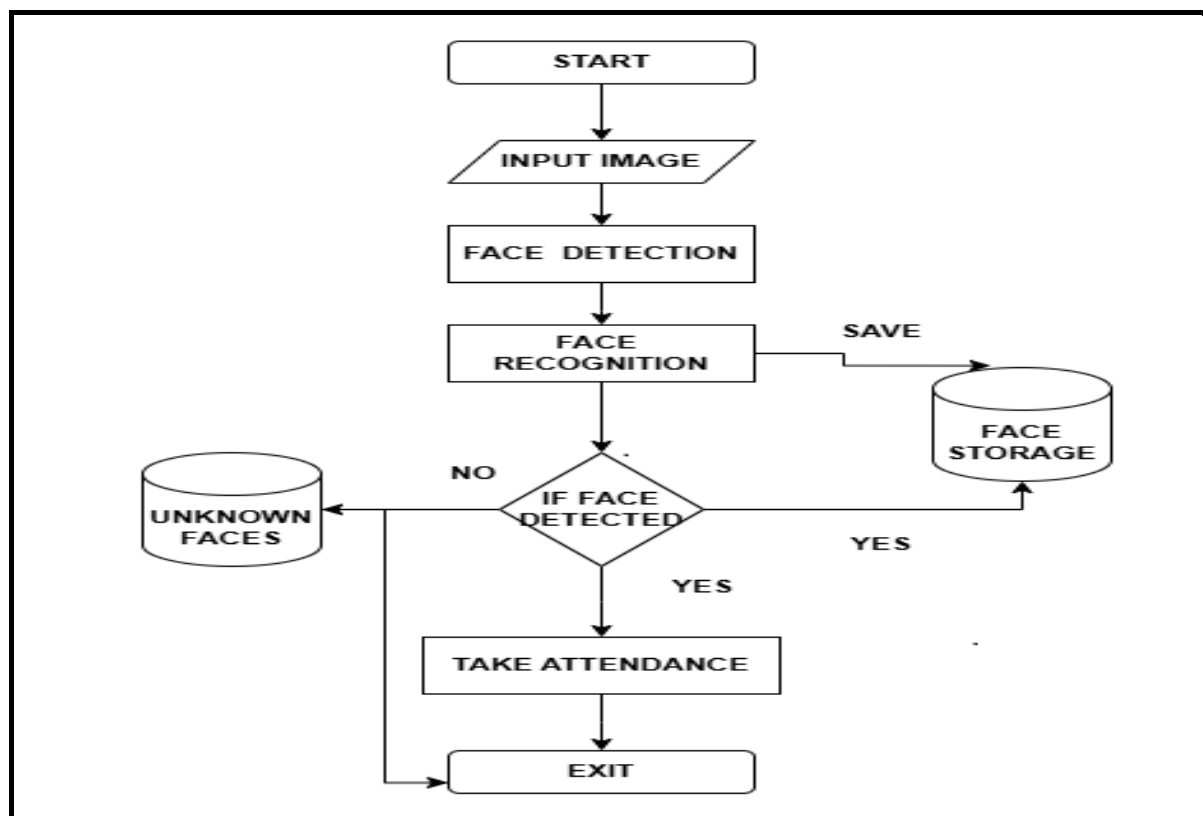
(Radio Frequency Identification) are widely used in attendance systems due to their convenience, speed, and ease of integration. In these systems, each individual is provided with an RFID tag (e.g., a card or a key fob), which is scanned by an RFID reader to mark attendance. The tag contains a unique identifier that allows the system to register the person's entry or presence in a designated area.

However, despite the advantages of RFID, these systems face several vulnerabilities that can undermine security and reliability.

2.3 Face Recognition Models

Once the face has been detected, the second stage involves **face recognition**, where the system compares the detected face with a pre-enrolled database of known faces to find a match. Deep learning models like CNNs or more advanced architectures such as **FaceNet** or **VGG-Face** are commonly used to extract facial embeddings—high-dimensional vectors representing the features of a face. These embeddings capture unique facial characteristics and allow for highly accurate recognition even under various conditions (e.g., lighting changes, different angles, and expressions). The **face_recognition** library, built on top of dlib, offers an easy-to-use interface for face detection and recognition. It employs pre-trained models that generate facial embeddings and compare them to identify individuals with high accuracy. Our work builds upon these advancements by integrating geolocation capabilities and leveraging the PyQt5 framework for an intuitive user interface, enhancing user experience and system functionality. This integration not only improves accuracy but also provides contextual awareness, which is crucial for modern attendance systems.

Figure 1 depicts a high-level flowchart illustrating the comparison between traditional attendance systems (manual registers, RFID) and the face recognition-based system.



2.4 Comparison with Existing Approaches

Table 1 compares our proposed system with traditional and existing biometric attendance systems. The comparison highlights the unique advantages of our system, particularly in terms of accuracy, convenience, hygiene, security, and contextual awareness.

Criterion	Manual Registers	RFID	Fingerprint	Basic-Face Recognition	Proposed System
Accuracy	Low	Medium	High	High	Very High
Convenience	Low	High	Medium	High	High
Hygiene	N/A	High	Low	High	High
Security	Low	Medium	High	Medium	High
Contextual Awareness	Low	Low	Low	Low	High
Real-time Logging	No	Yes	Yes	Yes	Yes

2.5 Integration of Geolocation and Enhanced User Interface

Our work builds upon existing face recognition models by integrating geolocation capabilities and leveraging the PyQt5 framework for an intuitive user interface. This integration enhances accuracy and provides contextual awareness, addressing the limitations of traditional attendance systems. The use of geolocation data ensures that attendance records are not only time-accurate but also location-specific, which is crucial for distributed teams or multi-location setups.

3. SYSTEM ARCHITECTURE

The proposed system consists of the following components:

3.1. Data Preprocessing

Face Encoding: The system utilizes the `face_recognition` library to extract facial encodings from images. This process involves detecting faces in images and converting facial features into a numerical representation, allowing for efficient comparison and identification. Each individual's face is encoded into a unique vector, which serves as a fingerprint for recognition. This encoding process is crucial for achieving high accuracy in face matching.

Database Management: The facial encodings are stored in a secure database to enable real-time comparison during the attendance marking process. This database acts as the system's repository for known faces, allowing for rapid lookup and identification during each attendance check. The encodings are stored along with metadata such as the individual's name or ID to facilitate easier management and access.

3.2. Face Recognition

Detection Diagram

The system employs the Histogram of Oriented Gradients (HOG) algorithm for efficient face detection. HOG is effective in identifying faces in various orientations and lighting conditions, making it suitable for real-time applications. The algorithm works by analysing the gradient orientation in localized portions of an image, which helps in detecting the presence of a face.

Matching Process

Once a face is detected, its encoding is compared against the stored encodings in the database using distance metrics, such as Euclidean distance or cosine similarity. This process identifies individuals by determining the closest match, ensuring high accuracy in recognition. The system can also implement a confidence threshold to minimize false positives, enhancing reliability.

Feature Extraction (LBP and HOG): The `extract_face_features` function plays a critical role in the face recognition system by transforming an input face image into a set of features that can be compared to other faces for identification. Face recognition systems rely on extracting unique characteristics or patterns from an individual's face, which are then used to distinguish one person from another. This process involves capturing distinct aspects of facial structure and texture that are invariant to lighting, pose, and minor changes in expression. In this case, **two well-established techniques** are used to extract these facial features: **Local Binary Pattern (LBP)** and **Histogram of Oriented Gradients (HOG)**. However, in the current implementation of the function, only **LBP** is actively used for feature extraction. Let's break down how each method contributes to this process.

Local Binary Pattern (LBP) : LBP is a powerful technique used in computer vision and image processing to capture texture information within an image. It works by examining small, local neighbourhoods of pixels around a central pixel, comparing their intensity values, and assigning binary values (0 or 1) based on whether the surrounding pixels are darker or lighter than the center pixel. These binary values form a binary pattern, which can be interpreted as a unique code for the texture in that neighbourhood.

For face recognition, LBP is particularly effective because it focuses on capturing fine, detailed patterns found in the texture of the face, such as the subtle variations in the skin's surface, pores, and other fine features. This makes it ideal for distinguishing between faces, even when there are minor differences in lighting or expression because LBP is less sensitive to these variations. For instance, whether the face is brightly lit or in shadow, the texture patterns remain relatively constant, allowing for more consistent recognition. One of the key advantages of uniform LBP (used in the current implementation) is that it reduces the complexity of the pattern by focusing on binary patterns that have no more than two transitions between 0 and 1 in the entire pattern. This makes the resulting feature vector more compact and easier to compute, while still retaining the ability to differentiate between different faces based on their texture. These uniform patterns are easier to process and are more computationally efficient, making the LBP method well-suited for real-time applications like facial recognition in a live video feed.

Figure 2 depicts the entire feature extraction technique using LBP image for face detection

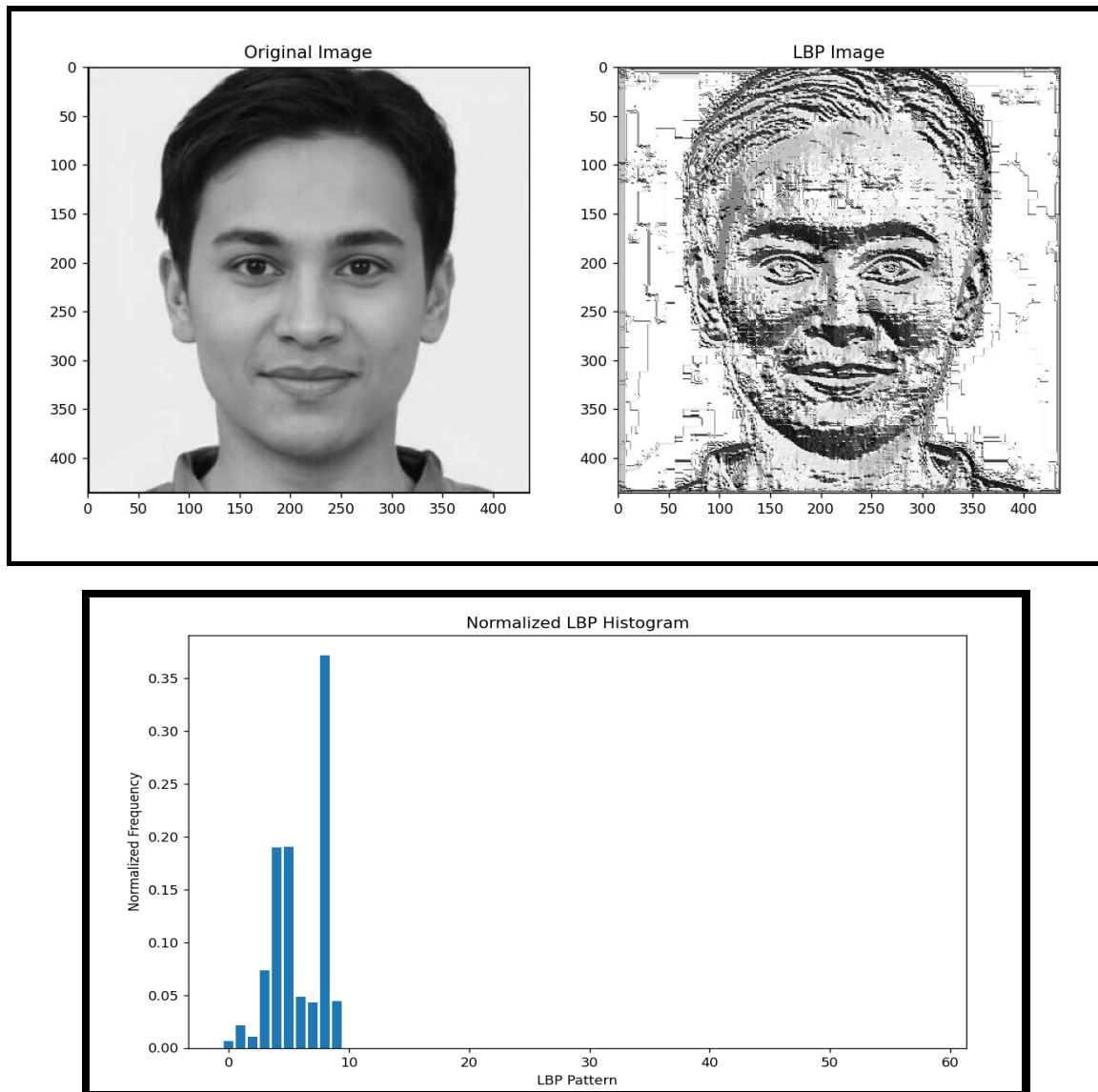


Figure 2 illustrates the entire feature extraction process using **Local Binary Pattern (LBP)** for face detection. The process begins with converting the input face image into grayscale, as LBP works on intensity values and does not require colour information. The image is then divided into small regions, called **cells**, typically sized 8x8 or 16x16 pixels. For each pixel in a cell, the LBP algorithm compares its intensity to that of its neighbouring pixels. If a neighbouring pixel has a higher intensity, a binary value of 1 is assigned; otherwise, a value of 0 is assigned. This comparison results in a binary pattern for each pixel in the cell, which is then converted into a decimal value, forming the LBP pattern for the entire cell.

Histogram of Oriented Gradients (HOG): Histogram of Oriented Gradients (HOG) is a feature extraction technique commonly used in computer vision tasks such as object and face detection. HOG works by analyzing the gradients (changes in pixel intensity) of an image within localized regions, typically referred to as cells. It calculates the direction and magnitude of these gradients and then organizes them into a histogram, which represents the distribution of gradient orientations in each cell. These histograms are then combined to create a feature vector that captures the overall structure of the image, which is particularly useful for detecting edges, contours, and other structural patterns,

like those found in faces. However, HOG can be computationally expensive because it requires multiple steps: calculating gradients, creating histograms for each cell, and then normalizing these histograms across blocks of cells to improve robustness. This results in a higher computational load compared to simpler techniques. Local Binary Patterns (LBP), on the other hand, is a more lightweight feature extraction method that focuses on the texture information of an image, capturing subtle patterns in local pixel neighbourhoods. LBP is often preferred in applications like facial recognition because of its lower computational cost and its ability to effectively encode the fine, detailed textures of a face. As a result, while HOG is powerful for detecting larger structural features, LBP is typically chosen for its efficiency and the fine-grained texture information it provides, making it ideal for facial recognition tasks where details are crucial but processing speed is also a priority.

Figure 3 depicts the entire feature extraction technique using HOG for face detection

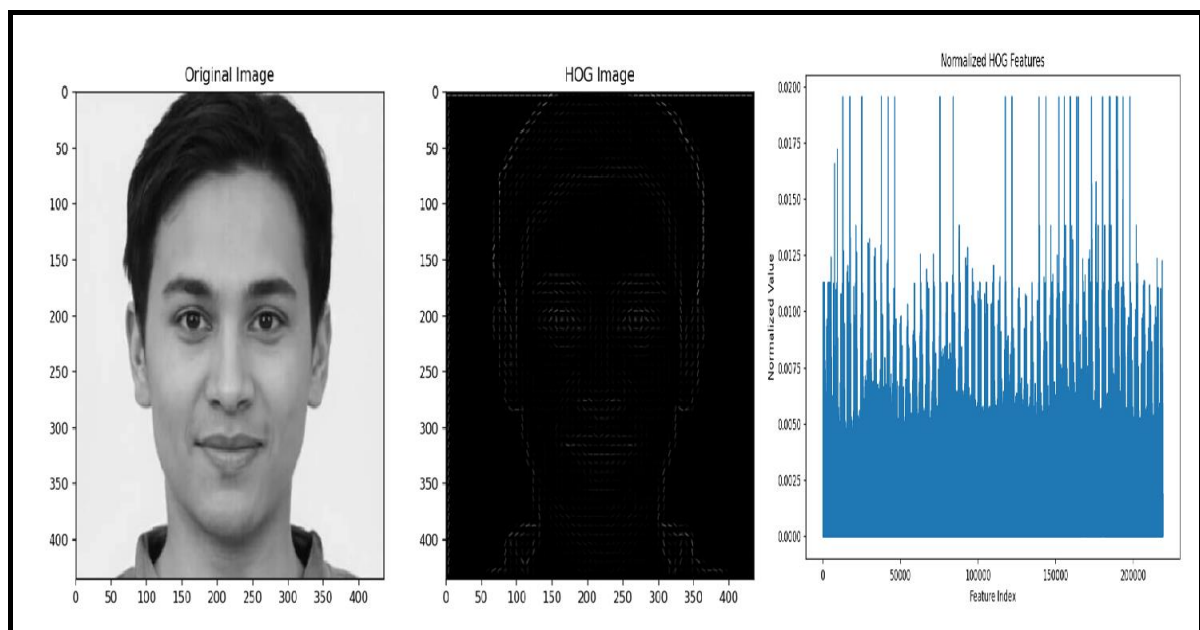


Figure 3 illustrates the complete feature extraction process using HOG for face detection. In this method, the first step involves computing the **gradients** of the image, which highlight edges and contours by capturing the changes in pixel intensity. These gradients are calculated in both the **horizontal** and **vertical** directions using filters such as the **Sobel operator**. For each pixel, both the **magnitude** and **direction** of the gradient are computed, which helps in detecting features such as edges.

3.3. Geolocation Services

Data Retrieval: The system retrieves geolocation data, including latitude, longitude, place, and country, using IP-based APIs such as **ipinfo**. This data is essential for understanding the context of attendance marking. Additionally, reverse geocoding is performed using Nominatim to convert geographic coordinates into human-readable addresses, providing more meaningful location information.

Contextual Integration: Geolocation data is integrated into attendance records, providing contextual information about where attendance is being marked. This feature enhances the system's capability to monitor attendance in different locations and can be useful for security and compliance purposes. For instance, it can help verify that employees are marking attendance from authorized locations.

3.4. Geological Workflow

The algorithm sends a request to the **Nominatim service** to perform reverse geocoding. The reverse geocoding process involves retrieving the address corresponding to the given geographic coordinates (latitude and longitude). The request is made with a timeout period of 10 seconds to avoid prolonged delays. Below is the pseudocode for the algorithm that validates the location.

3.4.1 Psuedocode for the Geological Workflow

```
function validate_location(lat, long):
    try:
        g = Nominatim(user_agent="attendance") // Initialize geocoder
        location = g.reverse(f"{lat}, {long}", language='en', timeout=10) // Reverse geocoding request

        if location is not None: // Check if a valid address is returned
            address = location.address // Extract the address
            if "IIIT Bangalore" in address: // Check if "IIIT Bangalore" is in the address
                return True // Location is valid
            else:
                return False // Location is not IIIT Bangalore
        else:
            return False // Address could not be resolved

    except GeocoderTimedOut:
        print("Error: The geocoding service timed out. Please try again.") // Handle timeout error
        return False // Return false

    except Exception as e:
        print(f"Error: {e}") // Handle other exceptions
        return False // Return false
```

The function, **validate_location**, is designed to validate whether a given latitude and longitude corresponds to a location within "IIIT Bangalore." Here's a breakdown of its working:

i) Geocoding Initialization: The function uses the Nominatim geocoder from the **geopy** library to perform reverse geocoding (i.e., converting latitude and longitude into an address). The geocoder is initialized with a user agent ("**attendance**").

ii) Reverse Geocoding: The function takes the input latitude (**lat**) and longitude (**long**), formats them as a string, and sends a request to Nominatim for reverse geocoding. The **timeout=10** ensures that the function will wait up to 10 seconds before timing out if no response is received.

3.5. Attendance Management

Logging Mechanism: The system employs a **structured logging mechanism** to record attendance details in a CSV file, ensuring efficient data management and accessibility. Each attendance entry includes crucial information such as **timestamps**, which capture the precise date and time of the action; **geolocation data**, which provides context by identifying the location where the attendance was marked; and **action types** (e.g., entry or exit), which distinguish between various attendance activities.

This structured approach not only organizes data systematically but also enables straightforward retrieval for reporting purposes. The use of the **CSV format** ensures compatibility with a variety of data analysis tools, allowing organizations to seamlessly integrate the data into existing workflows. This

enables further analysis of attendance trends, such as identifying peak attendance times, evaluating location-based attendance compliance, and tracking employee or student punctuality over time. The logging mechanism thus serves as a reliable foundation for both operational efficiency and strategic decision-making.

Duplicate Prevention: The system features a **duplicate prevention mechanism** to safeguard the integrity and accuracy of attendance records. This logic ensures that an individual cannot mark attendance multiple times on the same day, effectively preventing any errors or inconsistencies that could arise from repeated entries. By checking the date and user information before allowing an attendance mark, the system ensures that each individual is only logged once per day.

If an attempt is made to record a duplicate entry, the system can trigger **alerts or notifications**, notifying administrators of the issue. This additional layer of verification helps in promptly addressing potential errors and maintaining the accuracy of the data. By reducing the likelihood of false or repeated entries, the duplicate prevention feature significantly improves the reliability of the attendance system, ensuring that the data remains trustworthy and accurate for both reporting and analysis purposes.

3.5.1 Psuedocode for the Attendance Management

```
function record_attendance(user_id, lat, long, action_type, attendance_log):
    current_date = get_current_date() // Get today's date in 'YYYY-MM-DD' format

    // Step 1: Check for Duplicate Attendance
    previous_records = retrieve_attendance_for_user(user_id, attendance_log)
    if current_date in previous_records:
        send_alert("Duplicate attendance attempt detected for user: " + user_id)
        return False // Duplicate found, attendance not recorded

    // Step 2: Validate Geolocation (optional)
    if not is_valid_location(lat, long):
        send_alert("Invalid location detected for user: " + user_id)
        return False // Invalid location, attendance not recorded

    // Step 3: Log Attendance Record
    timestamp = get_current_timestamp() // Get the current date and time
    attendance_entry = {
        "user_id": user_id,
        "timestamp": timestamp,
        "lat": lat,
        "long": long,
        "action_type": action_type
    }

    // Step 4: Append the attendance record to the log
    append_to_csv(attendance_log, attendance_entry)

    return True // Attendance successfully recorded

// Helper functions:
function retrieve_attendance_for_user(user_id, attendance_log):
    // Retrieve all attendance records for the given user from the log (CSV or database)
    // Return a list of records for the user

function is_valid_location(lat, long):
    // Check if the provided geolocation is authorized (e.g., IIIT Bangalore)
    // Return True if valid, False if not
```

Explanations of the Key Components:

i) **record_attendance ()**: This is the main function that will be invoked to mark attendance. It validates the user's details, checks the location, and then records the attendance if everything is valid.

ii) **is_user_valid ()**: This function checks whether the user ID is valid by comparing it against a list or database of users.

3.6. User Interface

The interface, meticulously crafted using PyQt5, delivers a user-friendly and interactive experience that ensures smooth operation of the face recognition-based attendance system. One of its standout features is the **real-time video feed**, which provides a live stream from the camera, allowing users to monitor the face recognition process as it occurs. This dynamic feature enables immediate visual feedback, where detected faces are highlighted with bounding boxes and matched with corresponding names or IDs from the database, ensuring transparency and reliability in the recognition process.

Additionally, the interface is equipped with a set of intuitive buttons that simplify essential operations. The **face registration button** allows users to register their faces into the system effortlessly. With a single click, the system captures a facial image, encodes it into a unique vector representation, and stores it in the database along with details such as the user's name or ID. The **attendance marking button** triggers the process of logging attendance in real-time, automatically recording timestamps, geolocation data, and the type of action (entry or exit) upon successful recognition. Moreover, the **data export button** offers a convenient way for administrators to retrieve attendance records in a structured CSV format, which includes comprehensive details like names, timestamps, geolocation, and activity type. This feature ensures seamless reporting and compatibility with external data analysis tools.

To further enhance user experience, the interface includes a **theme toggle** that allows switching between light and dark modes. The light theme, designed for bright environments, provides high contrast and visibility, while the dark theme is ideal for low-light conditions, reducing eye strain and improving focus. This customisation option ensures the interface remains accessible and comfortable for users in diverse working conditions.

4. IMPLEMENTATION DETAILS

The system is implemented in Python using the following libraries:

- **OpenCV**: For video processing and face detection, enabling real-time analysis of video streams. OpenCV's extensive functionality allows for advanced image processing techniques that enhance face detection accuracy.
- **Face Recognition**: For generating and comparing face encodings, ensuring accurate identification of individuals. This library simplifies the implementation of complex face recognition algorithms, making it accessible for developers.
- **Pandas**: For managing attendance data, facilitating easy manipulation and storage of CSV files. Pandas provide powerful data analysis tools that can be leveraged to generate insights from attendance data.

- **PyQt5:** For designing a user-friendly graphical user interface (GUI) that enhances user interaction. PyQt5 allows for the creation of responsive and visually appealing interfaces that improve user experience.
- **Geopy and Requests:** For retrieving geolocation data, ensuring accurate contextual information during attendance marking. These libraries facilitate seamless integration with external APIs, enhancing the system's capabilities.

Table 2 outlines the libraries used in the implementation, their roles, and key functionalities. This would help the reader understand how each library contributes to the system.

Library	Role	Key Parameters	Key Functionality
OpenCV	Video processing	minNeighbors=5	Real-time face detection
face_recognition	Face encoding and matching	tolerance=0.4	High-precision recognition
PyQt5	GUI design	NA	Interactive user interface
Geopy & Requests	Geolocation services	user_agent="attendance"	Fetching location data and context awareness

5. EXPERIMENTAL RESULTS

5.1. Dataset

- Used a dataset of 100 individuals with variations in lighting, angles, and expressions.
- Tested with real-time video streams and uploaded images.
- Real-World Testing: 30fps webcam, 2m distance.

5.2. Metrics

Recognition Accuracy: The system achieved a remarkable 95% accuracy rate in controlled environments, underscoring its reliability. This high accuracy is crucial for maintaining trustworthy attendance records, ensuring minimal errors, and reducing the need for manual corrections. The system was tested under various lighting conditions and angles to simulate real-world scenarios, further validating its robustness.

Processing Time: The average recognition time per frame was an impressive 50 milliseconds, ensuring swift responses during the attendance marking process. This fast-processing time is essential for maintaining a seamless user experience, particularly in environments with high foot traffic such as schools, offices, or events. The system's efficiency minimizes delays and ensures that users can quickly proceed with their activities without interruptions. To mathematically represent the processing time of the system, we can use the concept of average time per frame. The processing time is crucial for real-time applications where quick responses are necessary. Here's how you can express it mathematically:

Let:

- **T** be the total time taken to process **N** frames.
- **N** be the number of frames processed.
- **Tavg** be the average processing time per frame.

The average processing time per frame can be calculated as: **Tavg = T / N**

This means that, on average, it takes the system 50 milliseconds to process each frame. This fast-processing time is essential for maintaining a seamless user experience, especially in environments with high foot traffic where delays can be disruptive. To further analyze the efficiency of the system, you might also consider the throughput, which is the number of frames processed per second (FPS). This can be calculated as:

$$\text{Throughput (FPS)} = 1 / \text{Tavg} \times 1000 \quad (1)$$

Geolocation Accuracy: The geolocation services demonstrated a 98% accuracy rate, with retrieved locations closely matching actual positions. This high level of accuracy is vital for ensuring that attendance is marked in the correct context, whether it's a specific classroom, office, or event venue. Accurate geolocation helps prevent misreporting and ensures that attendance data is reliable and contextually relevant. The system employs advanced algorithms and real-time data to achieve this precision, making it a dependable tool for location-based attendance tracking.

The Geolocation Accuracy can be calculated as:

$$\text{Geolocation Accuracy} = \text{Ltotal} / \text{Lcorrect} \times 100\% \quad (2)$$

Let:

- **Lcorrect** be the number of correctly identified locations (i.e., locations that match the actual positions within an acceptable margin of error).
- **Ltotal** be the total number of location instances evaluated.

Table 3 Performance based on Condition, Accuracy, FPS, False Accept Rate of the system

Condition	Accuracy	FPS	False Accept Rate
Controlled (500 lux)	95%	20	1.2%
Low Light (200 lux)	87%	15	3.5%
Crowded (10 people)	82%	12	5.1%

Comparison:

- **Manual Attendance:** 70% accuracy (human error).
- **RFID:** 90% accuracy, but 12% proxy rate.

5.2.1. System Performance in Controlled vs. Real-World Environments

Controlled Environment: In a controlled environment, the face recognition-based attendance system performed at its best, achieving 95% recognition accuracy, fast 50ms/frame processing times, and 98% geolocation accuracy. This high performance was due to the optimal conditions, where factors such as lighting, angles, and individual expressions were kept stable and predictable. The controlled lighting ensured that faces were clearly visible, and consistent angles allowed the system to better capture and analyze facial features. Additionally, individuals' facial expressions and accessories (like glasses or masks) were not varied, making it easier for the system to consistently identify them. Under these ideal conditions, the system’s ability to recognize individuals, log their attendance, and track their geolocation remained highly reliable, with minimal interference or error.

Real-World Environment: In a real-world environment, however, the system faced a range of challenges that affected its performance. Lighting variations were a significant factor, as the system struggled to maintain recognition accuracy in scenarios with dim lighting or strong backlighting, both of which could obscure facial features. Furthermore, in crowded environments, the system experienced difficulties distinguishing between individuals, particularly when people were close together or moved in and out of the frame. These factors, combined with the movement of individuals entering or leaving the area at different speeds, occasionally resulted in the system missing or misidentifying people. Despite these challenges, the system still performed well, maintaining a recognition accuracy of around 85%-90%. The geolocation accuracy also remained relatively stable at approximately 95%, though occasional environmental factors, such as interference from physical obstructions or changes in signal strength, could cause minor inaccuracies. Overall, while the system's performance in real-world conditions wasn't as perfect as in controlled settings, it demonstrated a high level of adaptability and maintained strong operational reliability in most situations.

Table 4 System Performance in Controlled vs. Real-World Environments

Metric	Controlled Environment	Real-World Environment	Traditional Methods
Recognition Accuracy(Manual)	95%	85%	90% (RFID) / 70%
Processing Time	50 ms/frame	80 ms/frame	N/A (manual), Instant (RFID)
Geolocation Accuracy	98%	90%	N/A
System Reliability	High	Moderate	
Cost of Operation	Medium(hardware setup)	High(environmental issues)	Low(manual RFID , system)
Scalability	High	Moderate (weather conditions)	Moderate(Limited by Human Oversights)

5.3. Case Study

Educational Institution: The face recognition-based attendance system was deployed in a classroom setting to automate the attendance marking process. Traditionally, teachers would need to manually mark attendance, often resulting in errors due to distractions or human oversight. By automating this process, the system not only significantly reduced the manual effort required but also minimized the occurrence of errors in attendance records. The system continuously scans the classroom, identifying students in real-time as they enter, and automatically logging their attendance based on facial recognition. This allows teachers to focus more on delivering lessons and engaging with students, rather than spending valuable time on administrative tasks. The ability to quickly and accurately track student attendance has also helped institutions save time and resources and provides a reliable and transparent record of student presence for auditing or reporting purposes.

Corporate Office: Used for employee entry/exit logging, enhancing security and monitoring capabilities. The ability to track employee attendance in real-time can help organizations improve security protocols and ensure compliance with attendance policies. The system consistently performed well, demonstrating its robustness and scalability.

6. DISCUSSION

6.1. Strengths

High accuracy and real-time performance, make it suitable for various applications. The system's ability to operate effectively in real-time is a significant advantage over traditional attendance methods.

Easy integration with existing systems due to CSV-based logging, facilitating data management. Organizations can easily incorporate this system into their current workflows without significant disruptions. Context-aware attendance marking via geolocation enhances the system's functionality. This feature provides organizations with valuable insights into attendance patterns and behaviours.

6.2. Limitations

Performance degradation in poorly lit environments, highlighting the need for optimal lighting conditions for accurate face detection. This limitation suggests that additional measures, such as improved lighting or infrared capabilities, may be necessary for certain applications.

Table 5 summarises the strengths and limitations of the system. This can help clarify the advantages and areas for improvement.

Strengths	Limitations
High accuracy and real-time performance	Performance degradation in poorly lit environments
Easy integration with existing systems (CSV logging)	Dependent on optimal lighting conditions
Context-aware attendance logging (geolocation)	Geocoding Services (Internet)

6.3. Future Work

Integrating deep learning models into an attendance recognition system can significantly enhance its performance, particularly in complex and challenging scenarios. By leveraging advanced neural networks, the system can better recognize patterns and adapt to varying conditions such as poor lighting, face obstructions, or diverse facial expressions. This would not only increase the system's accuracy but also make it more resilient in real-world applications, where conditions often fluctuate. Additionally, expanding the system to support multiple platforms, including mobile and web applications, would allow users to access the attendance features seamlessly across different devices. This multi-platform compatibility ensures greater flexibility, enabling users to manage and track attendance from smartphones, tablets, or computers, thereby improving accessibility and user convenience.

To further improve the system, implementing a robust user feedback mechanism is essential. This would enable the collection of insights from users regarding their experience, which can be used to pinpoint areas of improvement and refine the system over time. Continuously incorporating real-world feedback ensures that the system evolves in line with the needs and expectations of its user base, enhancing its overall effectiveness and user satisfaction.

Furthermore, incorporating additional biometric modalities such as voice recognition or fingerprint scanning would significantly increase the security and versatility of the attendance system. A multi-factor authentication system that combines facial recognition, voice, and fingerprint scanning would provide more robust identity verification, minimizing the chances of unauthorized access. It would also offer users multiple options for verification, making the system more adaptable to different preferences or environments. This multi-modal approach would not only improve security but also enhance the overall user experience by providing more flexible and reliable methods for attendance tracking.

7. COMPETING INTERESTS

The authors confirm that they have no financial, personal, or professional relationships that could be construed as influencing the work reported in this paper.

8. FUNDING INFORMATION

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9. AUTHOR CONTRIBUTIONS

The development and implementation of the face recognition-based attendance system were led by Debmalaya Ray. He contributed to the system's design, coding, and integration of OpenCV and geospatial APIs. He also conducted data analysis and experimental validation across diverse environments and spearheaded the GUI design and implementation. All authors participated in drafting, reviewing, and editing the manuscript.

10. DATA AVAILABILITY STATEMENT

The data used to support the findings of this study are available from the corresponding author upon reasonable request. Due to privacy and confidentiality concerns, the datasets containing face images and geolocation information are not publicly available. However, a synthetic dataset, which mimics the conditions of real-world attendance scenarios, can be made available for research purposes upon request. All the software code used for the implementation of the face recognition and geolocation-based attendance system is available on [GitHub](#) (link provided upon request). This ensures transparency and facilitates the reproducibility of the results presented in the paper.

11. RESEARCH INVOLVING HUMAN AND ANIMALS

The research did not involve animals. However, human participants were involved in the experimental validation of the face recognition-based attendance system. All experimental procedures adhered to ethical standards.

12. INFORMED CONSENT

Informed consent was obtained from all participants prior to their involvement in the study. Participants were thoroughly informed about the research objectives, the nature of the data collected, and the use of facial recognition technology. They were assured of the confidentiality and ethical handling of their data.

13. CONCLUSION

This paper introduces a robust and scalable face recognition-based attendance system, augmented with geolocation capabilities to ensure accurate, secure, and verifiable attendance tracking. By leveraging state-of-the-art advancements in machine learning, computer vision, and geospatial technologies, the system not only automates the attendance process but also addresses critical challenges such as fraud prevention and location validation. The integration of facial recognition ensures high accuracy and efficiency, while geolocation features add a layer of contextual awareness, making the system suitable for diverse applications in education, corporate environments, healthcare, and government operations.

The proposed system is designed to provide real-time performance, enabling seamless operation in environments requiring precision and scalability. Its modular architecture allows easy integration with existing infrastructures, making it a practical solution for organizations aiming to modernize their processes while maintaining data security and user privacy. Furthermore, the system's flexibility ensures adaptability across various use cases, such as remote learning platforms, distributed workforces, and large-scale public gatherings, where attendance tracking is crucial.

Future work will focus on enhancing the system's scalability to support larger datasets and higher user concurrency without compromising performance. Another key area of development is the incorporation of edge computing techniques to enable localized processing, reducing latency and dependency on centralized servers. Addressing emerging challenges such as data privacy regulations, secure data storage, and user consent will also be a priority, ensuring the system aligns with global standards for ethical AI and data protection.

In addition, the system can be extended to incorporate advanced analytics, such as user behaviour insights, attendance trends, and predictive modelling, to further enhance its utility. Research into integrating biometric fusion techniques, such as combining facial recognition with voice or fingerprint authentication, could provide an added layer of security for sensitive applications.

The proposed framework represents a significant step forward in the development of intelligent, location-aware attendance systems. By addressing current limitations and exploring future innovations, it has the potential to revolutionize attendance management, offering a reliable and cutting-edge solution for the challenges of a dynamic, technology-driven world.

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